

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 42

Attach a Forensic Volume and Analyze Snapshot Evidence

Amazon EBS | Snapshots | EC2 | Session Manager | Cloud Forensics

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Completed part two of the forensic snapshot lab by logging into the **SecurityAudit** account, creating a CloudFormation stack named **Forensics**, and using the snapshot shared in Project 41 as a forensic evidence source. A new EBS volume was created from the shared snapshot, tagged with **Name = CSI**, attached to the Forensics analysis instance at **/dev/sdf**, and mounted inside the operating system as the first partition of the attached drive. After mounting the partition to a **/forensics** directory, the file system was inspected and a hidden evidence file named **seatcecastronomy.txt** was found containing a YouTube URL that the user identified as the **No More Secrets** video.

SECURITY OUTCOME

This lab closes the acquisition-to-analysis loop: the compromised account produced a shared immutable snapshot, and the SecurityAudit account created a separate analysis volume from that snapshot without modifying the original evidence source.

Business Problem

In incident response, responders need to inspect compromised systems while minimizing changes to the original evidence. Project 41 created and shared a forensic snapshot of the compromised instance volume. This project shows the next step: create a new EBS volume from the shared snapshot, attach it to a separate forensics host, mount it, and search the file system for evidence.

Objectives

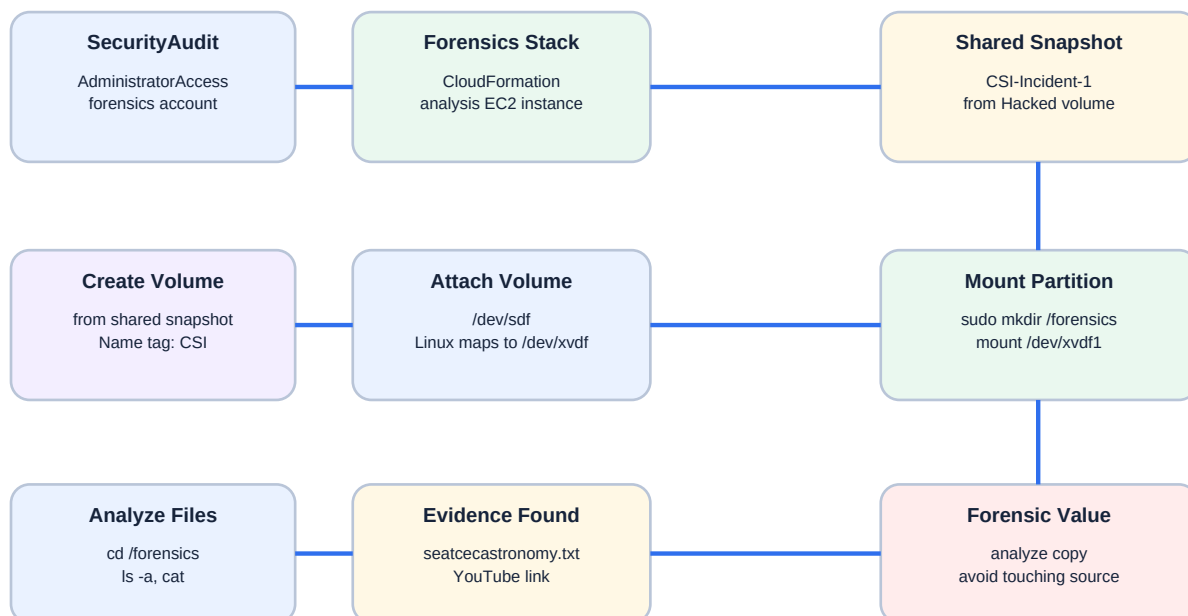
- * Sign into the SecurityAudit account with AdministratorAccess.
- * Create the Forensics CloudFormation stack using the provided lab template.
- * Open EC2 snapshots and locate the private snapshot shared from Project 41.
- * Create a new EBS volume from the shared snapshot.
- * Tag the new volume with Name = CSI.
- * Attach the CSI volume to the Forensics instance using device name **/dev/sdf**.
- * Connect to the Forensics instance using Session Manager.
- * Create a **/forensics** directory for mounting the evidence volume.
- * Mount the first partition of the attached drive to **/forensics**.
- * Inspect mounted directories and files using Linux command-line tools.
- * Identify the embedded evidence file and recover the YouTube link.

Environment

Cloud Provider	Amazon Web Services (AWS)

Primary Services	Amazon EC2, Amazon EBS, EBS Snapshots, AWS Systems Manager Session Manager, AWS CloudFormation
Account Context	SecurityAudit / AdministratorAccess
CloudFormation Stack	Forensics
Source Evidence	Shared EBS snapshot from Project 41 / CSI-Incident-1
Forensic Volume	EBS volume created from the shared snapshot
Volume Tag	Name = CSI
Attachment Target	Forensics EC2 analysis instance
Device Name Entered	/dev/sdf
Linux Device Mapping	Expected /dev/xvdf with first partition /dev/xvdf1
Mount Directory	/forensics
Mounted Partition	First partition of the attached drive
Evidence Found	seatcecastronomy.txt containing a YouTube URL
Status	Part 2 complete; cleanup not documented by user in this update

Architecture



The snapshot stays immutable; analysis happens on a new EBS volume attached to a separate forensics instance.

Figure 1 - Forensic volume analysis workflow using a shared EBS snapshot.

Implementation Summary

1. Signed into the SecurityAudit account with AdministratorAccess.
2. Created a CloudFormation stack named Forensics from the provided lab template.
3. Opened EC2 and navigated to Snapshots.
4. Selected the shared private snapshot created in the prior CSI lab.
5. Created a new EBS volume from that shared snapshot.
6. Added a tag with Key Name and Value CSI.
7. Attached the newly created CSI volume to the Forensics EC2 instance.
8. Selected /dev/sdf as the requested device name.
9. Connected to the Forensics instance using Systems Manager Session Manager.
10. Created a /forensics directory for analysis.
11. Mounted the first partition of the attached volume to /forensics.
12. Inspected directories and hidden files in the mounted file system.
13. Found seatcecastronomy.txt and used cat to reveal a YouTube URL.

Command Pattern

```
lsblk
sudo mkdir /forensics
sudo mount -t xfs /dev/xvdf1 /forensics
cd /forensics
ls
cd home/ec2-user
ls -a
cat seatcecastronomy.txt
```

The lab guide notes that the console device name can be entered as **/dev/sdf**, while newer Linux kernels may expose the attached volume internally as **/dev/xvdf**. The mounted partition used for analysis is **/dev/xvdf1**.

Evidence

Figure 2 - Forensic EBS Volume Created from Shared Snapshot

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state
☐	CSI	[REDACTED]	gp3	8 GiB	3000	125	[REDACTED]	[REDACTED]	[REDACTED]	usw2-az2 (us-west-2a)	Available

The EC2 Volumes evidence shows a volume tagged CSI, using gp3 storage, 8 GiB size, 3000 IOPS, 125 throughput, created in us-west-2a and available. Redactions protect volume and snapshot identifiers.

Figure 3 - CSI Volume Attached to Forensics Instance

Attach volume [Info](#)
Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
[REDACTED]

Availability Zone
usw2-az2 (us-west-2a)

Instance [Info](#)
[REDACTED]

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd[f-p] for data volumes.

ⓘ Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel Attach volume

The attach-volume evidence shows the selected volume and Forensics instance redacted, Availability Zone usw2-az2 / us-west-2a, and device name /dev/sdf. The AWS console warning notes that newer Linux kernels may rename devices internally to /dev/xvdf through /dev/xvdp.

Figure 4 - Embedded Evidence File Recovered from Mounted Volume

```
root@ [REDACTED] ec2-user]# ls -la
.. .bash_history .bash_logout .bash_profile .bashrc .ssh seatceastronomy.txt
root@ [REDACTED] ec2-user]# cat seatceastronomy
ash: catseatceastronomy: command not found
root@ [REDACTED] ec2-user]# cat seatceastronomy
at: seatceastronomy: No such file or directory
root@ [REDACTED] ec2-user]# cat seatceastronomy.txt
https://www.youtube.com/watch?v=jnuQmbZavzA
root@ [REDACTED] ec2-user]#
```

The terminal evidence shows the mounted file system being inspected from the ec2-user home directory. The hidden-file listing includes seatceastronomy.txt, and cat seatceastronomy.txt reveals a YouTube link.

Security Analysis

This lab demonstrates a practical forensic workflow in AWS: acquire evidence in one account, share the snapshot, and analyze a copy in a separate security account.

Creating an EBS volume from a snapshot does not modify the original snapshot. This is valuable because responders can repeatedly create clean analysis volumes if one is altered during investigation.

The snapshot and the analysis volume must be handled carefully. The volume created for analysis can be modified by the analyst, but the snapshot remains the preserved source copy.

Analysis in SecurityAudit reduces exposure to a compromised workload account and supports cleaner separation of duties.

Device names can be confusing in AWS because the requested device name and the Linux-visible device name can differ. The lab used /dev/sdf in the console and /dev/xvdf1 inside Linux.

Snapshot sharing remains a double-edged capability. It supports forensic transfer, but the same mechanism can be abused for data exfiltration when sharing permissions are too broad.

Lessons Learned

EBS snapshots can be used as immutable source evidence for forensic workflows.

Private snapshots shared into a security account can be converted into EBS volumes for analysis.

The forensic analysis volume must be created in the same Availability Zone as the analysis instance.

Attaching a volume is not enough; the operating system must recognize it and the correct partition must be mounted.

Linux commands such as `lsblk`, `mkdir`, `mount`, `cd`, `ls -a`, and `cat` are enough to perform a basic file-system inspection.

A separate forensic account is a cleaner place to perform investigation than the account where the incident occurred.

Production Improvements

- * Use a dedicated incident response or forensic acquisition account instead of a general SecurityAudit account.
- * Automate snapshot collection, sharing, tagging, and evidence tracking with Step Functions or incident-response automation.
- * Use encrypted snapshots and KMS key-sharing procedures for cross-account forensic transfer.
- * Record chain-of-custody metadata, including snapshot ID, source volume ID, account, region, timestamp, analyst, and case ID.
- * Mount forensic volumes read-only when possible, or create repeatable clean analysis volumes from the original snapshot.
- * Centralize CloudTrail evidence for `CreateSnapshot`, `ModifySnapshotAttribute`, `CreateVolume`, `AttachVolume`, and `DeleteVolume` events.
- * Apply guardrails to prevent accidental public snapshots while allowing approved forensic sharing.
- * Use hardened forensic AMIs with tools such as timeline generation, malware triage, file hashing, and artifact collection.

Skills Demonstrated

EBS snapshots	Create volume from snapshot	lsblk and mount
Cross-account evidence flow	Attach volume	Hidden file review
SecurityAudit analysis account	Session Manager	Forensic documentation

Resume Bullet

RESUME

Performed a cross-account AWS forensic workflow by creating an EBS analysis volume from a shared compromised-instance snapshot, attaching it to a SecurityAudit forensics instance, mounting the first partition, and recovering embedded evidence from the mounted file system.

STAR Interview Story

Situation: Part one of the lab produced a forensic snapshot of a compromised EC2 volume and shared it with the SecurityAudit account.

Task: Use the shared snapshot to perform analysis in a separate account without modifying the original compromised volume.

Action: I created the Forensics stack, located the shared private snapshot, created a CSI-tagged EBS volume from it, attached the volume to the Forensics instance as /dev/sdf, connected through Session Manager, mounted /dev/xvdf1 to /forensics, inspected the file system, and recovered the embedded seatceastronomy.txt link.

Result: The two-part workflow demonstrated forensic acquisition and analysis using EBS snapshots, cross-account sharing, a separate analysis instance, and Linux file-system inspection.

Version History

v1.0.0	July 30, 2026	Initial documented implementation. Logged into SecurityAudit, created Forensics stack, created CSI volume from shared snapshot, attached volume to Forensics instance as /dev/sdf, mounted first partition to /forensics, inspected the mounted file system, found embedded evidence file and YouTube link, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- * Cloud Security Lab a Week (S.L.A.W.) - Attach a Forensics Volume and Find the Prize!
- * Project 41 - Create a Forensic EBS Snapshot and Share It with SecurityAudit.
- * User-provided forensic volume creation evidence screenshot.
- * User-provided attach-volume evidence screenshot.
- * User-provided embedded secret terminal evidence screenshot.